

ALEX HILLMAN

Electrical & Computer Engineering Student

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SUMMARY

Final-year Electrical & Computer Engineering student at UCT with a love for embedded systems and design. Hoping to pursue a Master's in radar.

EDUCATION

University of Cape Town Feb 2023 – Dec 2026

BSc (Eng) in Electrical & Computer Engineering (Electrical and Electronics Engineering)

Rondebosch Boys' High School Jan 2016 – Dec 2020

National Senior Certificate

- Leadership: Prefect (Civic and Social Responsibility), SRC Secretary, Chairman of First Aid, Interact, Cabaret, Photography, and Civic and Social Responsibility societies
- Awards: Lesley Reeler Service Shield, GH Broster Trophy, Service Distinctions (Interact, First Aid), Sports Honours (Basketball)

EXPERIENCE

Course Tutor — University of Cape Town Feb 2026 – Present

- EEE3096S (Embedded Systems II) / EEE3097S (Engineering System Design) — Jul 2026 – Present
- EEE3088F (Design Principles) / EEE3089F (Electromagnetic Engineering) — Feb 2026 – Jul 2026

Event Manager — Butler's Pizza Call Dec 2024 – Present

City of Cape Town, Western Cape, South Africa · On-site

Back-up Manager — Butler's Pizza Call Jan 2024 – Present

Cape Town, Western Cape, South Africa

PROJECTS

CORRAL — Benthic Chamber Enclosure Redesign & Buoy Housing 2026

- Redesigned the enclosure subsystem for CORRAL, an autonomous dissolved-oxygen monitoring platform for the Langebaan Lagoon intertidal zone, as part of a four-person UCT engineering team
- Replaced a bolted lid requiring tools and several minutes to deploy with a tool-free bayonet closure and dual O-ring seal, validated by hand calculations and Fusion 360 FEA that caught a stiffness failure the hand calcs missed
- FDM-printed and field-tested the prototype: radial seal passed, face seal leaked; documented root causes and corrective actions for the next iteration

Quantifying Bioturbation-Driven Oxygen Flux in Permeable Sediments 2026

- Building on the CORRAL redesign, investigating the chamber's hydrodynamic performance in distinguishing diffusive from advective oxygen flux in permeable intertidal sediments, where standard chambers fail
- Characterizing pressure gradients under different stirrer configurations, boundary-layer thickness, and mixing efficiency, using diffusion-only and advection-enabled modes to isolate the advective flux component
- Planning field deployments in Langebaan Lagoon comparing oxygen fluxes across defaunated sediment, surface topography alone, live sand prawns, and relict burrows

TheShiftRosterBot — Multi-Tenant Scheduling Platform 2026

- Built and operate a production Telegram-based scheduling system for a delivery operation in Cape Town, coordinating shifts for 30+ drivers across two independently-configured branches
- Redesigned into a multi-tenant architecture (Python, SQLite, Raspberry Pi) when a second branch introduced different scheduling rules — one application serving multiple branches, each with its own database and ruleset, sharing a common engine
- Added tested migrations, a shared test suite, and deployment discipline; companion bots handle voucher redemption and end-of-shift closing checklists

Radar-Driven Interactive LED Cube 2026

- Designed the circuitry for a freeform-soldered 3×3×3 RGB LED cube controlled by an ESP32 and four 74HC595 shift registers, for an experimental art installation combining radar, music, and visual electronics
- Integrated a 24 GHz radar sensor over UART to track the user's position in real time and map it within the cube via multiplexed LED control
- Built this into an interactive movement-based game with score tracking, animations, and a firework sequence on winning

SKILLS

Embedded Systems / Hardware: STM32, ESP32, Verilog, FPGA design, Fusion 360, KiCad

Software: Python, MATLAB, C, Git, SQLite

Other: Team leadership, event/operations management